CSE423 | Group Project Report

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Section - 01 | Group - 8

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Game Report: "Curvy Snake"

"Curvy Snake" is a classic snake game built using Python's pygame library. In this game, the player controls a snake that moves around the screen and eats targets to increase its length. The game ends if the snake collides with the borders of the screen or its own body.

The game starts with an input display that asks the player to select the difficulty level. The player can choose from three levels - easy, medium, and hard - by pressing the corresponding number key on the keyboard. The game's difficulty level determines the speed of the snake, the size of the snake's head and body segments, and the size of the targets.

Once the player selects the difficulty level, the game starts. The screen is divided into three parts - the game area, the score display, and the border. The game area is where the snake and the targets move around. The score display shows the player's score, which increases every time the snake eats a target. The border is a rectangle that surrounds the game area and serves as a boundary for the snake.

The snake's movement is controlled by the player using the arrow keys on the keyboard. The snake moves in the direction of the arrow key pressed by the player. The snake's head is a circle, and its body segments are also circular. The size of the head and body segments depends on the game's difficulty level. The snake's length increases every time it eats a target, and a new body segment is added to its tail.

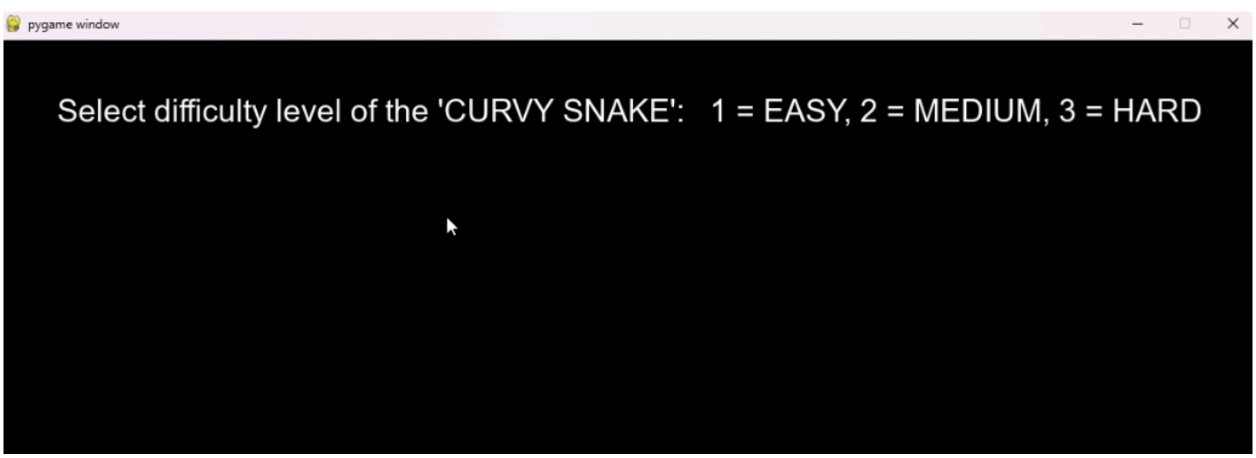
The targets are also circular and move around the game area randomly. The size of the targets depends on the game's difficulty level. When the snake's head collides with a target, the target disappears, and the snake's length increases. The player's score also increases by one.

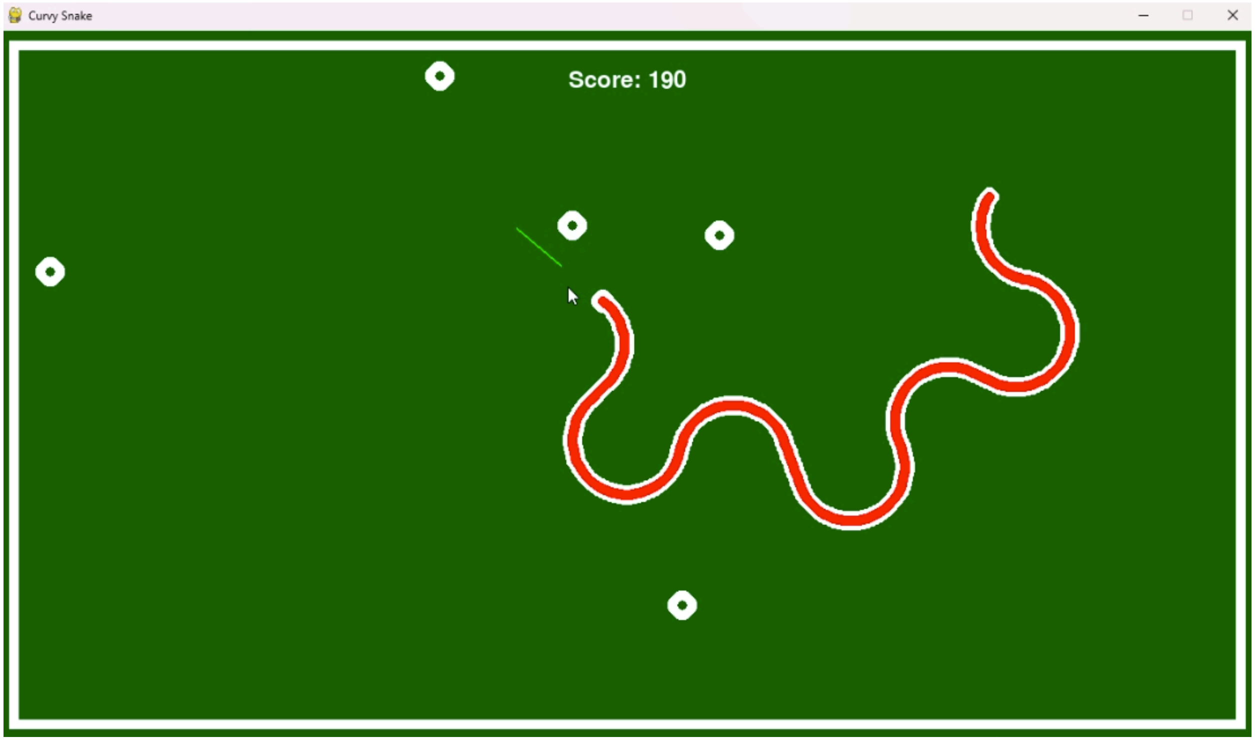
The game also includes several technical features, such as the mid-point line algorithm, the mid-point circle algorithm, and transformation. The mid-point line algorithm is used to draw a line between two points on the screen. The algorithm takes the coordinates of the two points as input and draws the line using the mid-point line algorithm. The mid-point circle algorithm is used to draw circles on the screen. The algorithm takes the center coordinates and the radius of the circle as input and draws the circle using the mid-point circle algorithm. The transformation is used to animate the eating animation when the snake eats a target. The animation increases the size of the target's circle and fades it out gradually.

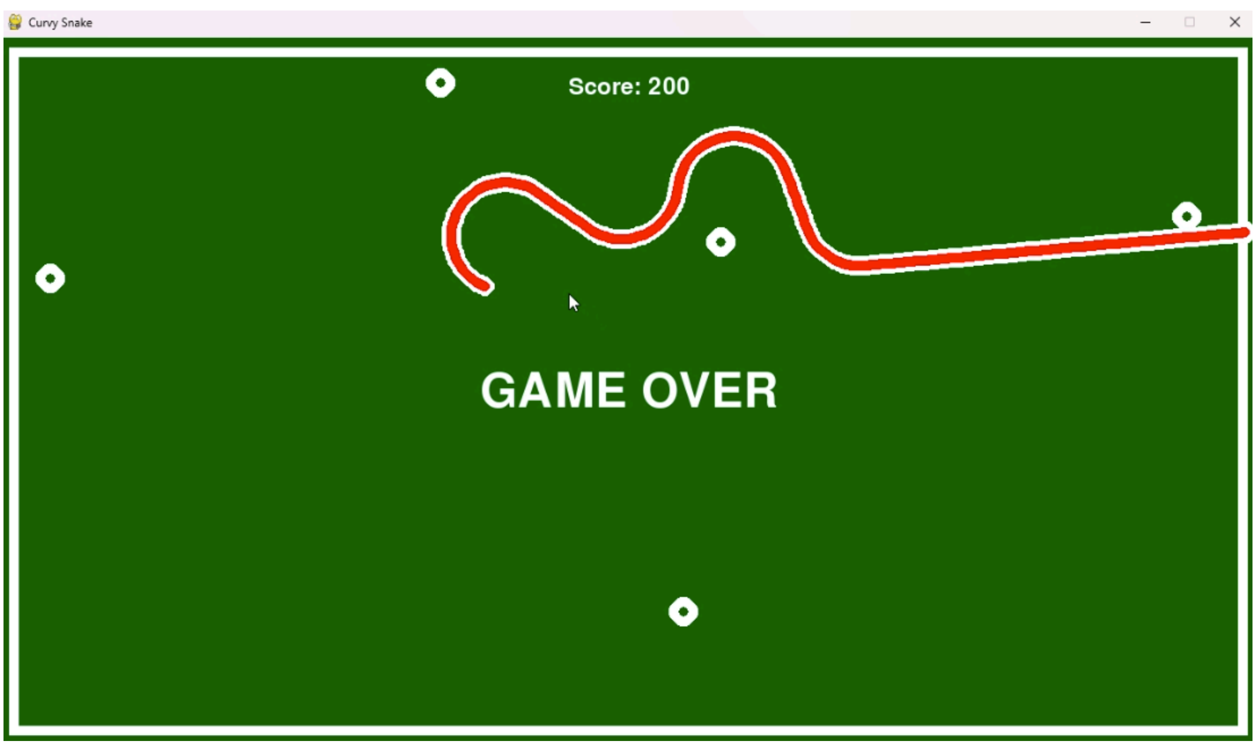
In terms of gameplay, the game is challenging and addictive. The snake's speed increases with the game's difficulty level, making it harder for the player to control it. The targets move around randomly, making it challenging for the player to catch them. The game's graphics are simple but effective, with the snake and the targets clearly visible on the screen. The game's sound effects are also appropriate, with a "chomp" sound played every time the snake eats a target.

In conclusion, "Curvy Snake" is a fun and challenging snake game built using Python's pygame library. The game's difficulty level, speed, and graphics make it a perfect game for players of all ages. The technical features of the game, such as the mid-point line algorithm, the mid-point circle algorithm, and transformation, add depth to the game and showcase the developer's skills.

Game screenshot:




Curvy Snake
Score: 190


Curvy Snake
Score: 200
GAME OVER